

SMART INDIA HACKATHON 2025



- **Problem Statement ID** – 25070
- **Problem Statement Title-** Secure Data
Wiping for Trustworthy IT Asset Recycling
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-** 94192
- **Team Name-** BYTE WIPERS

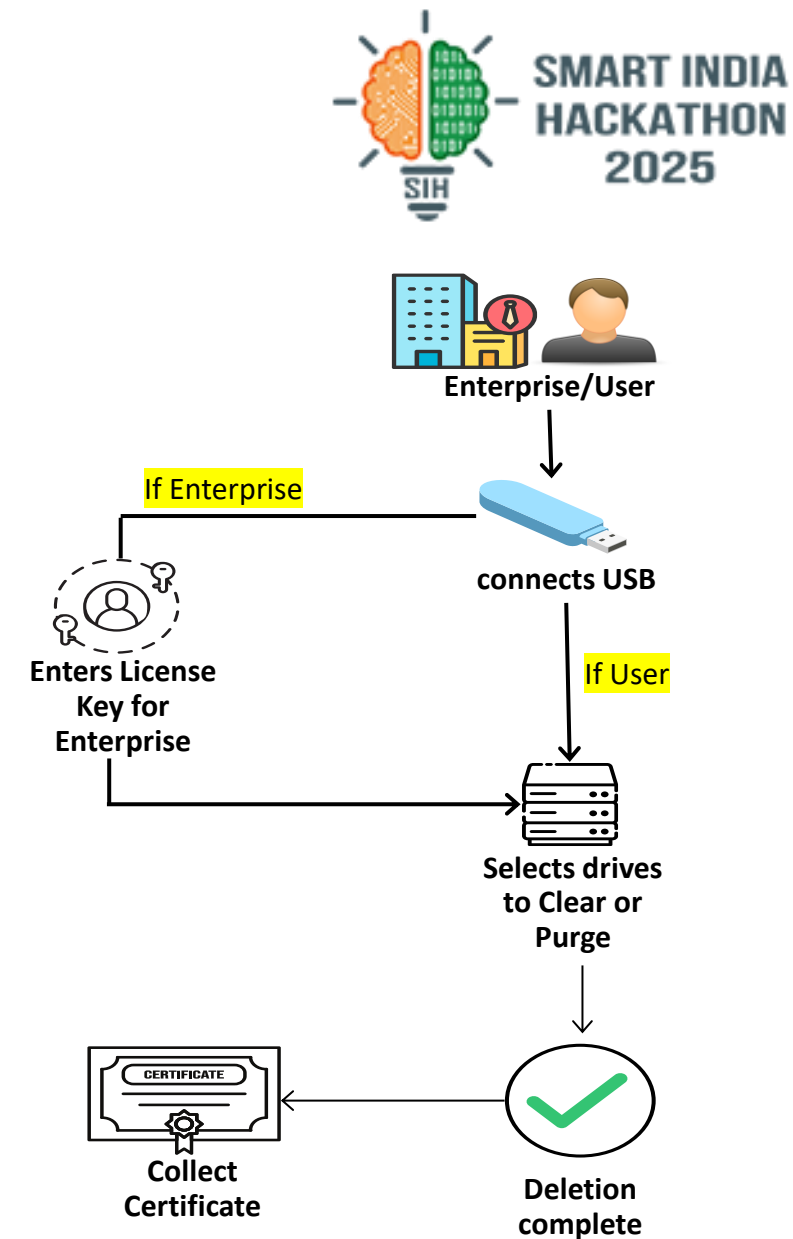


❖ **PROPOSED SOLUTION:**

Build trust in IT asset recycling by delivering a secure, user-friendly, and verifiable data wiping solution for **HDDs, SSDs**, and Android device.

Key Features:

- Standards-Compliant Security:** Implements secure data wiping aligned with **NIST SP 800-88** guidelines.
Covers HDDs, SSDs (including hidden sectors such as HPA/DCO), and Android device.
- Tamper-Proof Certification:** Generates **digitally** signed PDF and JSON certificates after each wipe. Certificates include device details, timestamp, and a unique **SHA-256** hash for authenticity.
- Blockchain-Verified Trust:** Certificates are anchored on the Hedera public ledger, ensuring immutability. Enables **third-party verification**, eliminating doubt or manipulation.
- One-Click Cross-Platform Application :** Provides a simple and intuitive interface for all users, including non-technical ones. Works seamlessly across **Windows, Linux**, and supports **offline bootable USBs**
- AI-Powered Assistant :** Wipable's **AI Agent** turns plain language into action fetching real compliance data, generating certificates, re-wiping devices, and giving clear wipe status for both **enterprises** and individuals.



Prototype Use case diagram

• Frontend:

• Web: NextJS, Shadcn, Material UI



• Desktop: Electronjs , NextJS, Talwind CSS



• Backend: Node JS, Express JS, Fast API, Clerk



• Blockchain: Hedera



• Database: Postgres, Redis , Chroma db



• Programming Languages: JavaScript, Rust,



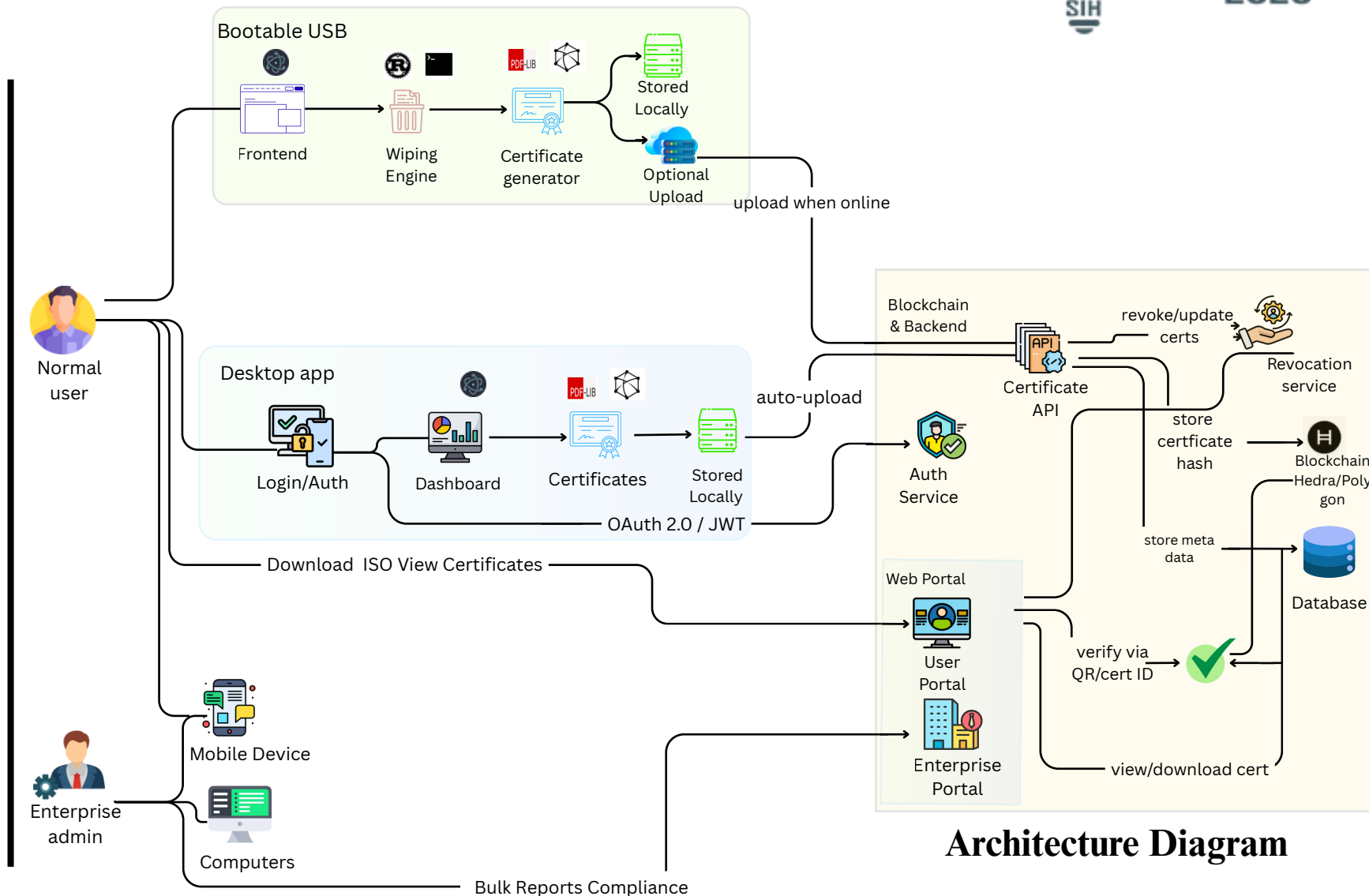
• Tools: Ventoy , Rufus



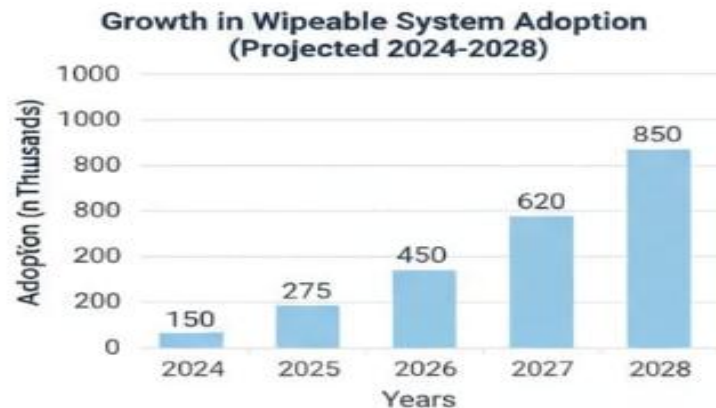
• Hardware: Bootable pen drive



• AI Agent:



Architecture Diagram



FEASIBILITY ANALYSIS :

- **Technical:** Secure wiping through ADB commands integrated with **Node.js & Electron**; supports both **desktop** and **Android apps**.
- **Financial:** Open-source foundation reduces licensing costs; scalable with minimal infrastructure.
- **Market:** High adoption in **enterprises**, refurbishing businesses, and **security**-conscious users.
- **Operational:** Easy to integrate into existing IT workflows with simple setup and **automation** support.



CHALLENGES & STRATEGIES :

- **Technical:** Ensuring wipe **verification** is reliable.
Strategy: Use checksum/logs to confirm data erasure.
- **Financial:** Scaling infrastructure for enterprise clients.
Strategy: Offer modular **SaaS** or on-prem **deployment**.
- **Market:** User trust in wiping accuracy.
Strategy: Provide compliance reports & certifications.
- **Operational:** Device compatibility variations.
Strategy: Build device compatibility **database** & updates.

Category	Potential Impact	Benefits of the Solution
Social	➤ Positive: Builds trust in recycling, protects privacy, promotes responsible disposal culture. ➤ Negative: Users may initially resist due to lack of awareness about certificate verification.	➤ Safer recycling of devices without fear of data leaks. ➤ Increased public confidence in digital security. ➤ Encourages wider adoption of IT recycling programmes.
Economical	➤ Positive: Unlocks hoarded IT assets worth ₹50,000+ crore, reduces costs of physical destruction, creates ITAD opportunities. ➤ Negative: Initial setup or infrastructure investment may be required for enterprises.	➤ New business opportunities for ITAD and refurbishing firms. ➤ Reduced long-term cost of secure disposal. ➤ Monetization of unused devices.
Environmental	➤ Positive: Reduction in e-waste through safe reuse and recycling; supports circular economy.	➤ Less landfill waste from unsafe disposal. ➤ Conserves natural resources by extending device lifecycle. ➤ Aligns with sustainability and green IT goals.

❖ REFERENCES

1. **NIST Special Publication 800-88 Rev. 1: Guidelines for Media Sanitization**
National Institute of Standards and Technology (NIST), USA.
<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-88r1.pdf>

2. **Blancco – Secure Data Erasure Whitepaper**
Explains enterprise-level data wiping standards and industry practices.
<https://www.blancco.com/resources/bp-physical-destruction-vs-secure-data-erasure/>

3. **Ministry of Electronics & IT (MeitY), Govt. of India – Guidelines on E-Waste Management**
Covers India’s e-waste problem and policies for safe disposal.
<https://www.meity.gov.in/content/e-waste>

❖ COMPARISON WITH EXISTING SYSTEMS

Feature / Tool	Our Solution	Blancco / BitRaser	DBAN / nwipe
NIST SP 800-88 Compliance	✓	✓	✗
SSD & Hidden Sector Support (HPA/DCO)	✓	✓	✗
Android Device Wiping	✓	✗	✗
Tamper-Proof PDF/JSON Certificates	✓	✓ (but private only)	✗
Blockchain Verification	✓	✗	✗
One-Click Easy UI	✓	✗ (enterprise-grade)	✗ (text-based)
Offline Bootable USB/ISO	✓	✓	✓
B2C Affordability	✓	✗ (high cost)	✓ (free, limited)